CHAPTER

33

Laser in Restorative Dentistry

Laser is being widely used in dentistry. The investigators have long been studying their effects on oral hard and soft tissues. Food and Drug Administration has approved the use of laser on dental tissues.

Laser is a device that transforms light of various frequencies into a chromatic radiation in the visible, infrared and ultraviolet regions with all the waves in a phase capable of mobilizing immense heat and power when focused at a close range.

Laser is acronym for '*light amplification by stimulated emission of radiation*'. The first laser was fabricated by Theodore Harold Maiman in 1960 using synthetic Ruby (aluminum oxide doped with chromium oxide). The light emitted was 107 times stronger than the solar light. The first lasers marketed for intraoral use were carbon dioxide lasers.

PRINCIPLE OF LASER

If a microwave beam containing photons passes through a cloud of ammonia gas, the gas molecules get raised to a higher energy state as the energy is absorbed. Earlier, Einstein observed that if a photon of correct size struck a molecule already in an excited state, that molecule would fall back to lower energy level and would emit a photon of exactly the same size and moving in the same direction as the entering photon. Thus, in case of ammonia gas, the molecules can undergo two possible changes, i.e. either moving to a higher state or being pushed back to a lower one. Normally, the first process would predominate, but if the majority of molecules are already in a higher energy state (termed as population inversion), then the released photon will speed on with the original photon. These two photons will strike two further molecules causing production of another two photons and a cascade reaction starts. This effect is known as 'Maser' (microwave amplification by stimulated emission of radiation). This principle was applied to the electromagnetic wavelength of light and the process known as 'LASER'.

LASER DEVICE

The laser device (Fig. 33.1) consists of the following:

a. ***Laser medium:*** Laser medium determines the wavelength of the light emitted from laser. The names of lasers are based on the active medium that is stimulated. The active medium can be a gas [(e.g. argon, carbon dioxide), a liquid (dyes)] or a solid crystal rod [e.g. neodymium yttrium aluminum garnet (Nd:YAG), erbium yttrium aluminum garnet (Er:YAG) or a semiconductor (diode lasers)].

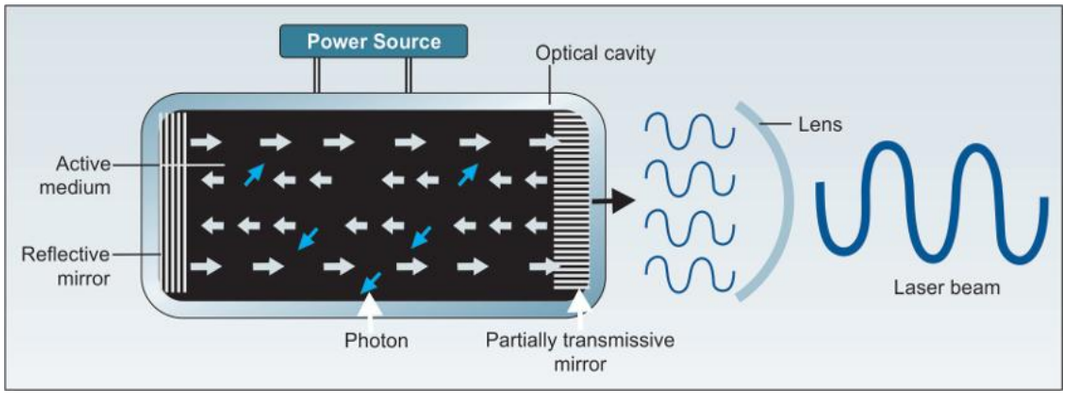


Fig. 33.1: The laser device

b. ***Optical cavity/laser tube/optical screen:*** It consists of two mirrors: One fully reflective and other partially transmissive, that are located on either ends of the optical cavity.

c. ***Power source:*** The power source excites or pumps the atoms of laser medium to their higher energy levels.

A population inversion happens when there are more atoms in the excited state rather than non-excited state. Atoms in excited state spontaneously emit photons of light, which bounce back and forth between the two mirrors in laser tube. As they bounce, they strike other atoms, stimulating more photons emissions. Photons as energy, wavelength and frequency, escape through transmissive mirror as a laser beam. The beam is directed via a flexible fiber-optic pipe/mirror, towards the point of application, where it is focused by a lens to the focal area. Energy impact that produces population inversion is called 'pumping'. Gas lasers are pumped by electrode discharge between electrodes. Laser is a beam of very high intensity light. It is monochromatic, coherent and collimated (directional).

- *Monochromatic:* Radiation in which the waves have same wavelength.
- *Coherent:* The waves are in a certain phase relationship to each other, both in space and time.
- *Collimated (directional):* The emitted waves have low divergence (nearly parallel).

Several characteristics differentiate laser light from ordinary light.

Ordinary light	*Laser light*
• Polychromatic	• Monochromatic
• Non-coherent	• Coherent
• Non-directional (scattered)	• Directional (minimum divergence)
• Low intensity	• High intensity

LASER DELIVERY SYSTEM

There are three delivery systems:

a. ***Articulated arms (with mirror at joints):*** For UV, visible and infrared lasers.

b. ***Flexible hollow waveguide:*** A flexible hollow waveguide or tube has an interior mirror finish. The laser energy is reflected along this tube and it exits through the handpiece at the surgical end with the beam striking the tissue in a non-contact fashion.

c. ***Glass fiberoptic cable:*** The third delivery system is a glass fiberoptic cable. This cable can be more pliant than the waveguide as it has a corresponding decrease in weight and resistance to movement, and is usually smaller in diameter (the delivery system of choice).

LASER EMISSION MODES

The dental laser device can emit the light energy in three modalities as a function of time:

a. ***Continuous ('constant on') emission:*** It is the continuous wave in which the beam is emitted at only one power level for as long as the operator depresses the foot switch.

b. ***Pulsed ('pushed on and off') emission:*** This is a pulse mode having periodic alterations of the laser energy, much like a blinking light. This mode is achieved by the opening and closing of a mechanical shutter in front of the beam pathway of a continuous wave emission.

c. ***Running pulse mode:*** In this mode, a large laser energy is emitted for an extremely short span; in microseconds followed by a relatively long span of laser (energy being the same).

The types of lasers and their mode of emission are summarized in Table 33.1.

Table 33.1: Types of lasers and their mode of emission

Type of laser	*Active medium (wavelength in nm)*	*Mode of emission*
Gas laser	• Argon (488, 515)	• Continuous
	• Helium-neon (633)	• Continuous
	• Carbon dioxide (10600, 10300, 9600)	• Continuous
Solid-state laser	• Nd:YAG (1064)	• Continuous/pulsed
	• Er:YAG (2940)	• Continuous/pulsed
	• Er:Cr:YAG (2780)	• Continuous/pulsed
	• Ho:YAG (2100)	
Semiconductor laser (diode)	• Gallium-arsenide (GaAs) (904)	• Continuous/pulsed
	• Gallium-aluminum-arsenide (GaAlAs) (780,890)	• Continuous/pulsed
	• Indium-gallium-aluminum phosphide (GaAlP) (630, 700)	• Continuous/pulsed

MECHANISM OF ACTION

The possible mechanisms of action of laser can be:

- *Photothermal ablation:* When high power lasers are used to vaporize/coagulate tissue through absorption (ablation).
- *Photo-mechanical ablation:* It causes the disruption of tissues due to shock wave formation, cavitations, etc.
- *Photo-chemical effects:* Use of light sensitive substances to treat cancer, etc.

LASER–TISSUE INTERACTION

This is the interaction of laser photons with the atoms/molecules of the target tissue. Lasers exhibit four types of interactions with the target tissue, depending on optical properties of the tissue.

The types of interactions are:

a. *Photobiological interactions:* The photobiological interactions are of four types:

i. *Reflection:* During reflection, the incident laser beam redirects itself without affecting the target tissues.

ii. *Absorption (absorption of laser energy by the intended target tissue):* The amount of energy that is absorbed by the tissue depends on the tissue characteristics, such as pigmentation and water content; and on the laser wavelength and emission mode. Argon has a high affinity for melanin and hemoglobin in soft tissue.

iii. *Transmission:* The transmission of laser energy through the tissues without affecting the same.

iv. *Scattering:* Scattering of the reflected light weakens the intended energy; may not produce any useful biologic effect.

b. *Photochemical interactions:* The basic principle of photochemical process is that specific wavelength of laser light is absorbed by naturally occurring chromophores, which are able to induce certain biochemical reactions. Photosensitive compounds, when exposed to laser energy, can produce a single oxygen radical for disinfection of the root canals.

c. *Photothermal interactions:* The radiant energy absorbed by tissue substances are transformed into heat energy; subsequently, produces the tissue effects (Table 33.2).

d. *Photomechanical and photoelectrical interactions:* These include photodisruption, photoplasmolysis and photoacoustic interactions. In photoacoustic effects, the pulse of laser energy on the dental tissues can produce a shock wave. When this shock wave explodes or pulverizes the tissue, it creates an abraded crater. Photoplasmolysis implies removal of tissues through formation of the electrically charged ions.

Table 33.2: Tissue effects with rise in temperature

Tissue temperature (°C)	*Observed effect*
<100°C	Water in soft and hard tissue boils producing explosive expansion
>100°C	Protein denaturation, hemolysis, coagulation, shrinkage
300°–400°C	Carbonization of organic material
400–1400°C	Inorganic constituents may melt and/or recrystallize; may vaporize

CLASSIFICATION

Lasers have been classified depending upon the following features:

1. According to wavelength:
   a. Excimer laser (wavelength 150–350 nm)
   b. Visible light laser (wavelength 350–750 nm)
      - Ruby laser
      - Argon laser
      - Helium-neon laser
   c. Infrared laser (wavelength 750–12000 nm)
      - Neodymium: Yttrium aluminum garnet (Nd:YAG)
      - Holmium: Yttrium aluminum garnet (Ho:YAG)
      - Erbium: Yttrium aluminum garnet (Er:YAG)
      - Carbon dioxide laser
2. According to the type of laser medium:
   a. Gas laser (argon, helium-neon, carbon dioxide)
   b. Solid-state laser (Nd:YAG, Er:YAG, Er:Cr:YAG, Ho:YAG)
   c. Semiconductor laser (gallium-arsenide, gallium-aluminum-arsenide, indium-gallium-aluminum-phosphide)
3. According to tissues to be lased:
   a. Soft tissue laser
   b. Hard tissue laser
4. According to potential hazards, lasers are classified according to American National Standard Institute's safety norms:
   - Class I: Do not pose a health hazard; inherently safe for eyes and skin (e.g. Laser printers).
   - Class II: Emit only visible light with low power output (1.0 mW) and do not normally pose a hazard because of the normal blinking (aversion reactions) of eye, e.g. surgical laser [continuous staring should be avoided; staring more than 10000 seconds produce burning in retina; e.g. commercial laser scanner (referred to as Class IIa)].
   - Class III:
     – Class IIIa: Any wavelength and output power less than 5 mW of visible light. It will not harm the unprotected eye within the blinking reflex

(one-fourth of a second) period (e.g. laser printers).
  - Class IIIb: Can produce a hazard to the unprotected eye, if viewed directly/from reflective light. Power output 500 mW (e.g. diode laser cutter).
- Class IV: Extremely hazardous from direct viewing, where the laser power is above 500 mW (e.g. lasers used in surgery; virtually all dental lasers).

This classification was accepted till 2002 and is designated as 'old classification'. The classification is modified, including concepts of '*maximum permissible exposure' (MPE)* (The highest power or energy density, i.e. W/cm^2 of a laser source that is considered safe) and the '*accessible emission limit' (AEL)* (the maximum power or energy, which can be emitted in a specified wavelength range and exposure time that passes at a specific distance).

The modified classification is:

- *Class 1:* Safe under all conditions; maximum permissible exposure (MPE) cannot be exceeded when viewing a laser with naked eye or with microscope.
- *Class 1M:* Safe under all conditions, except when passed through microscopes and telescopes.
- *Class 2:* Safe because of blink reflex; applies to visible light lasers (400–700 nm). Laser pointers and measuring instruments are in class 2.
- *Class 2M:* Safe because of blink reflex, if not viewed through optical instruments.
- *Class 3R:* Visible continuous lasers are limited to 5 mW (for pulsed lasers, this limit varies). 3R is considered safe with restricted beam viewing.
- *Class 3B:* 3B laser is hazardous, if exposed directly (direct viewing need protective eyewear). The accessible permissible emission limit (AEL) for continuum laser is 0.5W and wavelength 400–700 nm.
- *Class 4:* Lasers exceeding 3B level of AEL; mostly unsafe for eyes and skin.

CONVENTIONAL CATEGORIZATION OF LASERS

Lasers are generally categorized as of two types:

I. Soft tissue laser
II. Hard tissue laser

I. Soft Tissue Lasers

The soft tissue lasers are athermic, low energy lasers that generally utilize the semiconductor laser diodes (Figs 33.2 and 33.3). The wavelength of such lasers stimulates cellular activity. They aid in tissue regeneration and wound healing by increasing collagen production by fibroblast stimulation.

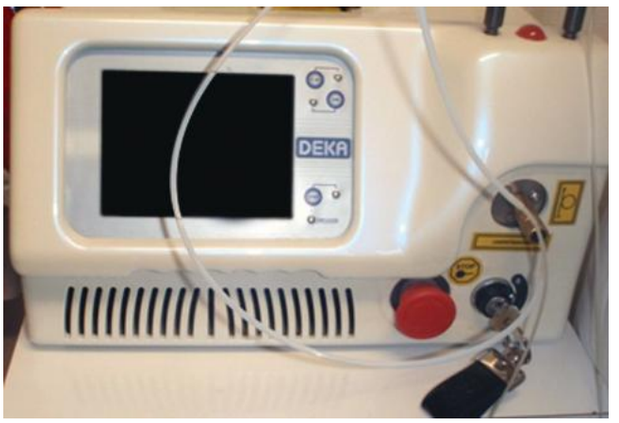


Fig. 33.2: Soft laser unit (DEKA)



Fig. 33.3: Soft laser unit (doctor smile)

Soft lasers are as follows:

- Helium-neon (He-Ne) laser
- Argon laser
- Gallium-arsenide (Ga-Ar) laser
- Gallium-aluminum-arsenide (GaAlAs) laser
- Carbon dioxide (CO_2) laser

II. Hard Tissue Lasers

The hard tissue lasers are thermic and of high energy, that are utilized in surgery as precise energy source, i.e. cut, coagulate and vaporize (Fig. 33.4).

Hard lasers are as follows:

- Erbium: Yttrium aluminum garnet (Er: YAG) laser
- Carbon dioxide (CO_2) laser
- Neodymium/yttrium aluminum garnet (Nd: YAG) laser
- Excimer lasers



Fig. 33.4: Hard laser unit

Advantages
• Conservative and precise restorative procedures. • Minimum use of sutures during surgical procedures; minimal bleeding. • Minimum damage to surrounding tissues. • Less traumatic to patients (convenient); maximum patient co-operation. • Multiple restorations can easily be carried out. • No vibrations, no smell, etc. • Since anesthesia is not used, the implications associated with anesthesia are minimized. • The contact mode is short; the operator can neatly visualize the preparation. • Less chances of mechanical trauma with use of lasers. • Lasers reduce bacterial counts; sterilize the field as well. • No adverse pulpal reactions. • Better healing; tissue regeneration. • Effectively removes caries, disinfect the area and provide clean surface for restoration. • Lasers can seal dentinal tubules, reducing the possibility of post-restorative sensitivity. • Restorative time is reduced.

Disadvantages
• Adequate cavity preparation not possible (drilling required for final shape). • Cannot be used on teeth that already had filling. • Cannot be used to repair existing amalgam fillings; vaporize mercury, making it highly toxic. • May damage pulp. • Inability to remove metallic and porcelain restorations. • Cannot be used to prepare crowns; not even to remove crowns. • Treatment being expensive; not cost-effective.

ROUTINELY USED LASERS

a. Carbon Dioxide (CO_2) Laser

Carbon dioxide (CO_2) laser (Fig. 33.5) is considered suitable for soft tissue surgeries, sealing of pit and fissures, prevention of caries, erosive lesions and bleaching etc. CO_2 is mainly used for intraoral soft tissue surgery, such as, frenectomy or removal of hyperplasic tissue (high degree of absorption of CO_2 wavelength by oral mucosal tissues; 90% water). Absorption of this energy results in cellular rupture and vaporization (photothermal effect). During this process, heat is generated and conducted into surrounding tissues; may damage the surrounding tissues. One advantage of this thermal damage is providing excellent hemostasis.

In case, control of the extent of lateral thermal damage is required, the laser application is to 'pulse' the laser wavelength (faster the pulse, the less time would be available for conduction into adjacent tissues).

Although CO_2 laser is primarily absorbed by water, it is also well absorbed by hydroxyapatite (component of enamel). The wavelength has the higher absorption in hydroxyapatite out of all other lasers (use as pit and fissure sealant/caries prevention). However, generation of high temperature is not conducive with hard tissue preparations (tissues adjacent to surgical site must be shielded from laser beam).

b. Argon Laser

Argon laser uses argon gas as the active medium. Energy passing through the gas generates coherent visible light with primary wavelengths of 488 and 514.55 nm. These visible blue-green laser light wavelengths have peak absorption in red pigment, such as, hemoglobin; thus, laser light is well absorbed in pigmented tissues with abundance of hemoglobin,



Fig. 33.5: CO_2 laser unit

hemosiderin and melanin. Argon laser light is not well absorbed by enamel and dentin or other non-pigmented tissues. These characteristics make argon laser energy very useful for cutting, vaporizing, coagulating and providing hemostasis on gingiva and oral mucosa. Argon laser is primarily used for root planing and curettage, gingival retraction, gingivectomy/gingivoplasty, frenectomy and treatment of oral lesions, etc.

c. Neodymium: Yttrium Aluminum-Garnet (Nd:YAG) Laser

Nd:YAG laser system emits its pulsed energy at 1064 nm; the energy is directed through a 320 μ silica fiber, using the high peak of free running pulse emission. The clinical applications are cutting and coagulation of soft tissues. It has an affinity for pigmented tissues; also minimizes the heat build-up in tissues.

d. Erbium: Yttrium Aluminum Garnet (Er: YAG) Laser

Er:YAG laser system, with wavelength of 2.94 μm, which coincides with absorption peak of water, results in good absorption in all biologic tissues, including enamel and dentin. Er:YAG laser does not induce any thermal changes. Water has a very high absorption for Er:YAG laser light. The incident laser radiation is absorbed in a thin surface layer causing sudden heating and vaporization of water. A high steam pressure leads to microexplosions with erupting particles. Because the tissue is not vaporized completely but only disintegrated into fragments, radiant energy is converted into ablation that alters the morphological structure of tissue. Er:YAG laser can be used for the removal of healthy hard tissue as well as carious decay without causing any thermal injury to adjacent tissues. Restorative materials, like composites and cements, can be ablated without using mechanical bur. Gold crowns and cast fillings cannot be removed because the laser beam is reflected by metals. No ablation effect is observed on ceramics. This laser is recommended for osteotomy, cyst removal and apicoectomy because of bone healing properties.

The thermal side effects are greatly reduced with Er:YAG laser. However, this laser beam is not easily transmitted through optical fibers and, therefore, is usually applied via mirror and arm delivery systems. A root canal preparation is not possible due to lack of suitable delivery system (Fig. 33.6A, B).

EXCIMER LASER

The following limitations of CO_2 and Nd:YAG laser, which led to the development of excimer laser, are:

- Vaporization and carbonization of dental tissue.



Fig. 33.6A: Er:YAG laser unit



Fig. 33.6B: Er:YAG laser unit (screen)

Advantages

- Thermal effects are minimal [Temperature rise is minimum (50%)].
- Prepare cavities in dental hard tissue without side effects.
- Bacteriological effect on prepared surface
- Possibility of tissue identification in selective removal of affected tissues.
- Zone of necrosis is small; no residual debris.
- No carcinogenic effect.

Disadvantage

- Expensive

Clinical use of lasers: Key points

- Always use lower power setting
- Operating area should be kept wet
- If patient feels pain or becomes uncomfortable, modify laser application as:
  - Pulses per second are decreased (energy is decreased)
  - Cutting tip is changed from contact mode to a non-cutting mode
- The sound of ablation (popping sound) from various tooth tissues is characteristic of laser

Table 33.3: Common lasers, their wavelength and clinical applications

Medium	*Wavelength (nm)*	*Enamel characteristics*	*Clinical application*
CO_2	10,600	• Roughness • Partially fused crystallizes • Charring • Cracking	• Resin bonding • Enamel etching • Caries reduction
Argon	514	• Reflection produces minimal effect on tissues	• Resin polymerization • Bleaching
Er:YAG	2,900	• Etched, scaly or flaky (rough surface)	• Cavitation • Enamel etching
Nd:YAG	1,064	• Reflection of beam is seen, until photoabsorptive is placed	• Decay removal • Etching
Ruby	693	• Cratering • Irregular surface (charring)	• Resin polymerization
Er, Cr:YSGG	2780	• Hard tissue removal	• Gingiva ablation
ArF	193	• Slight etching, smooth (lacking in charring)	• Cavitation • Enamel etching
Diode	980	• Surface tissue removal	• Photostimulation • Bleaching
KTP	532	• Surface tissue removal	• Bleaching

- Thermal side effect, which may damage pulp and periodontal tissue.
- Caries removal leads to structural changes, such as cracks, zones of necrosis, etc.

Excimers work by the process of ablative photo decomposition, which implies bond breaking of molecules with minimal thermal side effects (Temperature increase is approximately 12°C). Residual energy is converted into kinetic energy by expansion of residual gaseous phase. Excimers are emitted in UV spectral range.

The preparation of dental hard tissue with excimer lasers has been tried.

The available excimer lasers are:

- ArF 193 nm
- KrF 248 nm
- XeCl 308 nm
- XeF 351 nm

The common lasers, their wavelength and clinical applications are summarized in Table 33.3.

USES OF LASER IN RESTORATIVE (CONSERVATIVE) DENTISTRY

The high intensity lasers (HIL) are used for ablation, vaporization, cutting and coagulation of tissue; increase in temperature during work. The low intensity lasers (LIT) are used in reducing inflammation (promoting healing) and in cases where analgesia is required.

The way laser interacts with tissue depends upon optical characteristics of laser (wavelength, power, timings, etc.) and also on optical characteristics of target tissues (chromophores, fat tissue, etc.).

In restorative dentistry, high intensity lasers (Nd:YAG, Er:YAG, Er, Cr:YSGG, power diode, etc.) are used for cavity preparation, final removal of caries, internal conditioning of ceramics, conditioning of enamel and dentin, prevention of erosion, etc. Low intensity lasers are preferred in reducing microbial contamination after cavity preparation, dentin sensitivity, bleaching and also in maintaining periodontal health.

The main areas of usage of lasers in restorative dentistry are:

a. Tooth surface conditioning (Excimers, CO_2, Nd:YAG, Er:YAG)
b. Caries prevention (pit and fissure sealants) (CO_2, Nd:YAG, Excimers)
c. Cavity preparation (CO_2, Nd:YAG, Excimers, Er:YAG)
d. Caries detection (Argon)
e. Dentin bonding and curing (Argon, CO_2, Nd:YAG)
f. Bleaching (CO_2, Argon)
g. Dentin hypersensitivity (CO_2, Nd:YAG, Er:YAG)
h. Gingival care (Argon, He: Ne)
i. Prevention of erosion (CO_2)
j. Miscellaneous

a. Tooth Surface Conditioning

CO_2, Er:YAG, Er, Cr:YSGG, Nd:YAG lasers are being used for etching of enamel. CO_2 laser is preferred because of its favorable absorption in dental hard tissue (Fig. 33.7). In addition, CO_2 laser can be placed



Fig. 33.7: Etching with laser

at localized area when used with appropriate delivery system. This results in etching to a specific area of enamel; whereas, acid gels usually overextend the area.

Er:YAG (2.94 μm) and Er, Cr:YSGG (2.78 μm) have high affinity for water molecules in hydroxyapatite; subsequently, effectively absorbed by dental hard tissues.

The difference of composition of mineralized tissues in enamel and dentin results in lower ablation threshold for dentin as compared to enamel (Dentin has higher organic content; ablation of dentin requires lower energy as compared to enamel).

One of the main advantages of high energy laser is its antimicrobial action (Ablation of hard tissues is accompanied by microbial reduction of adjacent tissues).

A black dye is placed on enamel surface to increase absorption of laser beam. Total clinical time is reduced (seven seconds for conditioning and fifteen seconds for acid etching), as there is no need of washing or drying.

b. Caries Prevention (Pit and Fissure Sealants)

Dental caries is characterized by subsurface enamel demineralization. Laser energy has the potential to alter the enamel to make it less susceptible to this subsurface demineralization.

CO_2, Nd:YAG and excimers are commonly used.

CO_2 laser is suitable since its long wavelength can easily be absorbed by enamel.

Only the shallow depth is affected, thereby minimizing possible harmful effects on the pulp. The most effective wavelength for inhibiting artificial caries was found to be 9.3 to 10.6 μm.

Nd:YAG laser, when used as pit and fissure sealant, removes inorganic and organic debris in pit and fissure without injuring the surrounding healthy enamel. The laser effect would block the pits/fissures.

The important features of lasers helping caries prevention are:

i. Minimum energy is required to avoid damaging the dental pulp.
ii. Ability to direct the laser beam within the restricted area by means of a flexible beam guide.

Various hypotheses were put forward explaining how lasers prevent caries.

Earlier authors opined that Nd:YAG laser may increase the acid resistance of tooth enamel; melting thin surface layer of enamel.

The use of laser for the prevention of dental caries is based on reduction of micro-organisms, due to action on plaque and also morphological alteration of the enamel surface.

Nelson et al (1978) observed reduction of enamel permeability and also reduction of enamel solubility. Formation of pyrophosphate on lased enamel reducing enamel solubility was also suggested.

Oho and Morio (1990) observed that the lased enamel was found to have a high positive birefringence, which might be due to presence of micropores. The loss of water, carbonate and organic substances from enamel might have caused formation of micropores.

The micropores were thought to act as sites for deposition of ions released by an acid attack. During the treatment with lactate buffer, acid solution penetrates the enamel and calcium ions are released by acid decalcification. In unlased enamel, released ions diffuse to the surrounding surface. On the other hand in lased enamel, some of released ions would be trapped in microspaces, which lead to remineralization; responsible for changes of birefringence.

It is established that laser acts in the following ways:

- The enamel surface gets sealed and becomes less permeable to diffusion by ions into and out of enamel during demineralization.
- The composition of enamel is altered (reduced carbonate content of apatite crystals), which reduces its solubility and permeability.

The application of laser is most effective when the emitted laser light is matched to the absorption of target tissue. The degree of difference between these two wavelengths determines the amount of laser energy, i.e. reflected or absorbed into the target tissue; for example, when applied to enamel and dentin, treatment of tooth is different. Enamel and dentin have different peak absorption values of 9.6 μm and 2.9 μm, respectively. As a result, no single laser has proven fully versatile when applied to oral tissues. The routinely available lasers operate at one predetermined wavelength that cannot be changed to accommodate the absorption values of the treated teeth. The transmission wavelength of Er:YAG laser is equal to absorption of water (2.94 μm). The new generation laser [Mark III Free Electron Laser (FEL)]

Table 33.4: Effect on enamel using various laser types

Laser type	*Effect on enamel*
Carbon dioxide (10,600 nm)	• Increase in enamel resistance to acids in pits and fissures • Preventive effect is significant when combined with fluorides • Laser activation improved the photoactivated sealants • Only laser application may not be effective
Nd:YAG (1064 nm)	• Effective in increasing the acid resistance of enamel surface • Removes debris from pits and fissures; facilitate fluoride uptake • May not be effective in deep fissures
Er:YAG (2900 nm)	• Laser application lead to melting and re-crystallization of enamel surface • Increase enamel resistance to acid dissolution

can be adjusted in the range of 1.4–10 μm, which enhances its area of application.

Table 33.4 summarizes effects on enamel using various laser types.

c. Cavity Preparation

Lasers, when used for cavity preparations, usually produce too much heat, melt the enamel and damage the pulp.

CO_2 and Nd:YAG lasers have been used to remove carious dentin without causing damaging pulp.

Er:YAG laser has shown promising results. It is absorbed by water and hydroxyapatite, which partially accounts for its efficiency in cutting enamel and dentin.

When tooth tissue is vaporized by laser, its surface becomes darkened due to carbonization. This can be removed by applying a steady stream of water while simultaneously lasing the area.

Due to alteration of surface structure, the lased tooth becomes resistant to decay. In case of root caries, the placement of restorative material may not be necessary after laser application.

It is suggested that laser should be pulsed at intervals that are shorter than the stimulation period required to trigger nerve pulse transmission. When cutting the vital tooth, it can be lased for about 10 seconds (Fig. 33.8A to C).

The currently used lasers use pulse duration in seconds or now in microseconds. Even with these pulse durations, residual temperature may increase and cause undesirable thermal changes in underlying tissues.

Advantages

- Laser-induced analgesia (no need for local anesthesia).
- Less post-restorative sensitivity.
- Access to subgingival caries is improved; lasers vaporize gingival tissues.

Laser cavity preparation: Guiding steps

- Use of rubber dam; adequate isolation
- Gently touching the laser tip with target tissue
- Create adequate and effective ablation
- Constantly move laser tip over the target surface; achieve uniform cutting
- Move laser tip, as required for proper cavity depth
- Continual water spray to target tissue; achieve effective cooling
- Complete cavity preparation; usually irregular in shape
- Prepared cavity is restored with appropriate materials

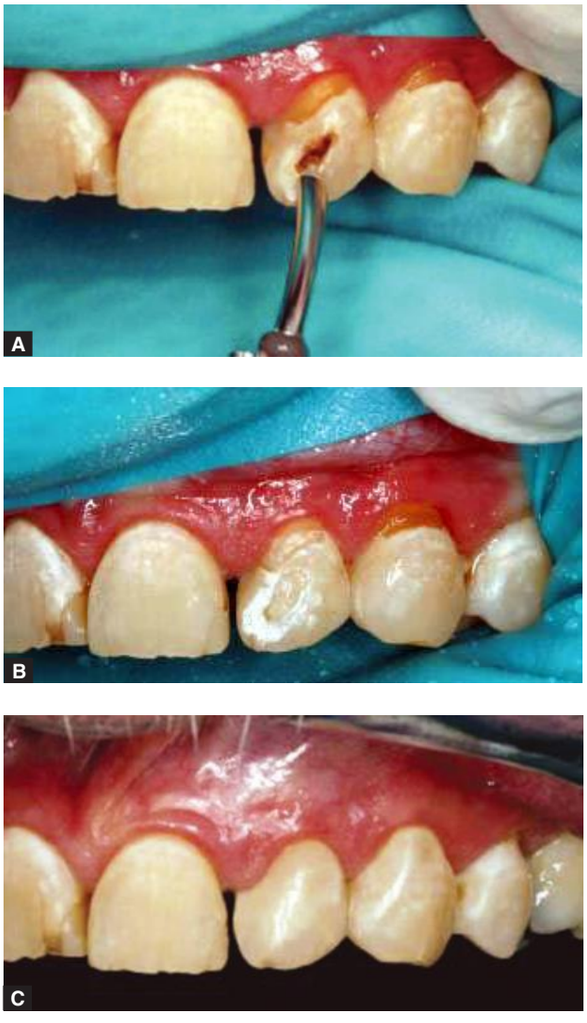


Fig. 33.8: Cavity preparation with Er:YAG laser: **(A)** Cavity preparation on maxillary lateral incisor; **(B)** Prepared cavity; **(C)** Restored cavity

Recently, ultrashort pulse lasers emit photons with pulse duration in pico/fem seconds (one-tenth or fifteenth of seconds). Cavities prepared with these modalities are well defined; still use is under practical consideration.

It is established that Er:YAG laser was equivalent to high speed drill in its ability to make cavity preparations in enamel and dentin. Er:YAG offer great advantage when used to prepare cavities (ablade hard tissues). Erbium laser produces no vibrations as there is no need of laser tip to contact the tooth structure (vibratory noise of conventional handpiece is 100–120 dB, depending upon intensity of vibrations). Erbium lasers are preferred for tooth preparation; 1.0 mm clearance is adequate between laser tip and tooth structure. However, cavity preparation with laser is not convenient. The conventional cavity preparation with angled wall or sharp edges are difficult to be adequately performed with high intensity lasers. The main cavity is to be prepared with conventional rotary instruments; the final caries can be removed with low intensity laser (caries removal and microbial reduction without damaging the pulp).

Laser-assisted cavity preparation
• Laser is used in up-down movement • Cutting at the tip direction • Cavity surfaces are rough; aid in adhesive restorations • Smear layer is removed • Patient movement may not hinder preparation

Conventional cavity preparation
• Used all along the tissue • Mostly side cutting • Cavity surfaces are smooth • Burs produce smear layer • May not be safe during patient movement

d. Caries Detection

It is established that intact enamel is luminous when exposed to argon laser at a wavelength of 488 nm. The carious lesions appear as dark areas. Proximal lesions may be visualized using laser before demineralization could be evident on radiographs.

DiagnoDent is a low pulsed caries detector. It exhibits 90% accuracy in diagnosing caries as compared with conventional diagnostic modalities. This device also has limitations because depth of penetration of light is limited to about 2.0 mm. Another drawback is that it only detects occlusal lesion and cannot identify interproximal lesions.

e. Dentin Bonding and Curing

It is established that dentin bonding is substantially increased (up to 300%), if the dentin is pretreated with CO_2 laser prior to bonding. Improved dentin-bonding with argon/Nd:YAG lasers has also been reported. The enamel surface to receive composite can be laser etched to provide the desired roughness. After restoration, the restoration–tooth interface is laser treated to reduce the marginal gap; subsequently, the recurrent caries.

Advantages
• Shorter curing time • Better physical properties • Increased depth of cure • Better polymerization • Reduced polymerization shrinkage

Lasers have been used to cure composites. The laser used is argon with a wavelength of 488 nm. This is near the wavelength of initiator used (camphorquinone) in composite resins.

f. Bleaching

Two types of laser are usually used (Argon laser emits a visible blue light, and CO_2 laser emits invisible infrared light).

The laser energy rapidly decomposes H_2O_2 to O_2 and H_2O. Argon laser is preferred for removal of dark stains than CO_2 laser.

g. Dentin Hypersensitivity

Laser has provided another treatment option for dentin hypersensitivity (Fig. 33.9).

Various types of lasers used are:

- He:Ne laser (632 nm)
- CO_2 laser (960 nm)
- Nd:YAG laser (1064 nm)
- Er:YAG laser (2940 nm)
- Er,Cr:YSGG (2780 nm)

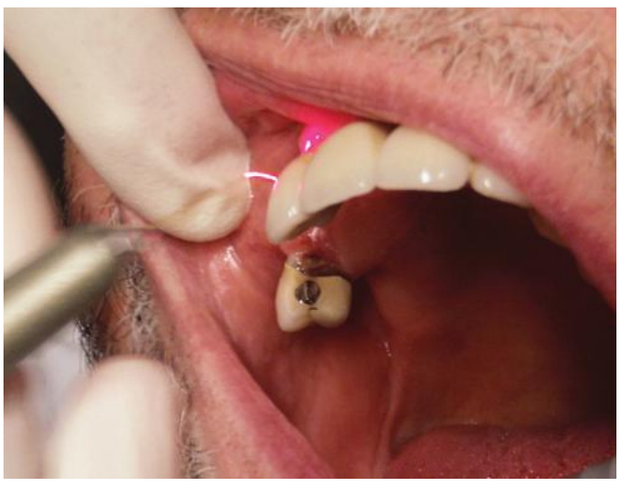


Fig. 33.9: Desensitization using laser

Lasers occlude the dentinal tubules by producing local changes around the exposed dentin. Low level laser therapy induces alterations within the nerve transmission of dental pulp, instead of altering the exposed dentin surface. Nd:YAG, Er,Cr:YSGG and CO_2 lasers have the ability to melt peritubular dentin and occlude dentinal tubules; subsequently, reducing hypersensitivity.

h. Gingival Care

The health of gingiva is mandatory during and after restorative procedures. Low intensity lasers are effective in maintaining and recovering healthy gingiva. The anti-inflammatory effect of low intensity lasers is due to increase in microcirculation, resulting in alteration in hydrostatic pressure reducing resorption. It has also shown effective tissue regeneration in gingiva. A multiple application may be required to achieve the desired results.

The anti-inflammatory and biomodulatory effects of lasers in gingival recovery are:

- Increase in fibroblasts
- Increase in endothelial cell proliferation
- Increase in collagen synthesis; better collagen fiber organization
- Increase in vascular flow
- Increase in growth factors
- Increase in leukocyte phagocytosis
- Decrease in crevicular fluid
- Decrease in MMP8 (metalloproteins)

i. Prevention of Erosion

Dental erosion refers to pathological loss of tissue without the involvement of bacteria. Acidic drinks are considered as the main causative factors, which may be difficult to avoid, being associated with habits.

Carbon dioxide lasers have been mostly tried to prevent erosion, due to its interaction with hydroxyapatite. The erosive preventive effects of carbon dioxide laser (10.6 μm, 5 second pulse, 0.3 J/cm^2 density energy) have been confirmed.

j. Miscellaneous

Laser can be used in variety of other fields such as:

- Nd:YAG laser is used for welding.
- CO_2, Nd:YAG, and Argon lasers are used to sterilize instruments and to kill bacteria on culture media, glass slides, etc.
- Exposure of dentin to laser leads to activation of dentinogenesis.
- Laser can produce 3-D images of objects. These images are called holograms. The study casts, etc., can be saved as holograms.
- Cavity treated with laser offers better adaptation of glass-ionomer cements.
- Laser produces effective analgesia; replace the need for local anesthesia.
- Laser is effective in soft tissue management.
- Incision and drainage of abscess
- Removal/softening of gutta-percha
- Effective in biopsy excisions
- Effective in vital pulp therapy (pulp capping)
- Effective in detecting vertical root fractures.
- Lasers are used to alleviate pain in temporo-mandibular joint.
- Laser optical tomography allows to see inside the tooth.
- Lasers help regenerate blood vessels and nerves.
- Laser energy is utilized for implant bed preparation; osteointegration of titanium implants.
- Laser-assisted biomimetic process is used to coat calcium phosphate over resin composite to create cementum-like osteoconductive surface.

LASER HAZARDS

The beneficial effects of lasers have been accepted in day-to-day practice; however, continuous use may have certain hazards.

The main hazards are as follows:

a. Effect on eyes (ocular damage)
b. Effect on skin and mucous membrane
c. Effect on enamel and dentin
d. Environmental/respiratory hazards
e. Electrical hazards (shocks)
f. Fire and combustion

a. Effect on Eyes (Ocular Damage)

Ocular damage (injury to the eye) can either be by direct emission from laser or by reflection from a specular surface. The primary ocular injury that may result from a laser accident is a retinal or corneal damage. Retinal injury is possible with emissions in the visible (400–780 nm) and near infrared (780–1400 nm) wavelengths (Table 33.5). It may cause 'scotoma' (loss of vision in the path of visual field; blind spot).

b. Effect on Skin and Mucous Membrane

Laser may damage skin and mucous membrane as a result of thermal interaction of radiant energy with tissue proteins. Temperature elevation of 21°C above normal body temperature can produce cell destruction by denaturation of cellular enzyme and structural proteins, which interrupts basic metabolic processes.

Factors affecting tissue damage are:

- Relative absorption and transmission of particular wavelength.

Table 33.5: Laser wavelength and pathological effect on eyes

Wavelength (nm)	*Effect on eyes*
180–315 (ultraviolet)	Inflammation of the cornea (like sunburn)
315–400 (ultraviolet)	Clouding of eye lens
400–780 (visible)	Retinal burn (photochemical damage to retina)
780–1400 (near infrared)	Retinal burn; cataract
1400–3000 (infrared)	Cataract, corneal burn
3000 and above	Corneal burn

- Pulse duration and pulse repetition rate.
- Level of radiation exposure.
- Relative degree of vascularity of tissue.

c. Effect on Enamel and Dentin

The deleterious effects of lasers on enamel and dentin are:

- Enamel exhibits gross craters (0.1–1.0 mm deep), depending on amount of energy delivered to the target area. In deeper penetration, dark speckling of exposed dentin can be seen. Examination under polarized light may show crystallographic changes.
- Dentin shows shallow and irregular craters (0.1 mm deep). Three distinct zones of dentinal destruction are: Central zone of complete dentinal destruction; an immediate surrounding area of partial dentinal destruction; and a scattered zone of dark speckling beyond first two zones.

d. Environmental/Respiratory Hazards

Inhaled air-borne contaminants can be emitted in the form of smoke or fume generated through thermal effect of laser on tissue or through accidental escape of toxic chemical and gases from laser itself.

Lasers damage functioning of the respiratory system. It is established that detectable level of ethanol, isopropanol, etc. found in operating laser room are potentially damaging to the respiratory system. Toxic gases/chemicals (fluorine, etc.) are a common hazard in dental operatory.

Lasers and electrocautery are used to vaporize, coagulate and cut tissues. The vapors, smoke and particulate debris produced during these surgical procedures are called *laser plumes*. The plume contains particles with mean diameters of 0.1–0.3 µm. Within this plume of carbonized tissue, viable tumor cells and viral particles have been cultured. Laser plume may contain dust, viruses, bio-aerosols, etc., which can be hazardous to the staff working in laser clinic. These may lead to eyes, nose and throat irritation.

e. Electrical Hazards (Shocks)

Because lasers use high currents and high voltage power supplies, these are potentially hazardous. Electrical hazards of lasers can be grouped as electrical shock hazards, electrical fire hazards or explosion hazards.

An utmost care should be exercised in installing and using laser equipment; should be performed by a qualified person.

f. Fire and Combustion

Laser can be hazardous, if flammable materials are lying alongside the use. Flammable solids (papers, cloths, etc.), liquids (acetone, ethanol, etc.) and gases (nitrous oxide, aromatic fragrance, etc.) can be ignited, if exposed to laser beam.

LASER SAFETY

It is established that even small amount of laser light can lead to eye injuries. Laser safety designs utilize use of laser in such a way, so as to minimize the risk of laser accidents, especially involving skin and eyes. American National Standard Institute (ANSI) provides control measures for laser hazards. The maximum permissible exposure (MPE) limit should strictly be looked into prior to use of lasers.

ANSI has provided guidance for the use of lasers in a series, ANSI 136. The standard protocol is:

136.1: Safe use of lasers
136.2: Safe use of optical fibers and LED sources
136.3: Safe use of lasers in health care
136.4: Laser safety measurements for hazard evaluation
136.5: Safe use of lasers in educational institutions
136.6: Safe use of lasers outdoors
136.7: Labelling laser protective equipment
136.8: Lasers in research and development
136.9: Lasers in manufacturing environment

Safety Measures

Everyone uses laser should be aware of the risks involved. The routine complacency dealing with invisible risks can lead to health hazards, which should be avoided following safety measures.

American National Standard Institute (ANSI) has classified lasers as per their safety parameters. ANSI has specified 'maximum permissible exposure', i.e. the level of radiation to which a person may be exposed without hazardous effects on adverse biologic changes. In some applications, open beams are required making it necessary to define an area of potentially hazardous laser radiation. This is *normal hazard zone* (NHZ), which is defined as *"a space within*

which the level of direct, scattered or reflected laser exceeds the maximum permissible exposure".

The safety requirements include:

- A laser warning signal outside the clinic.
- Use of barriers within the operatory.
- Use of eyewear to protect against reflected laser light or accidental direct exposure.
- The residue left after tissue ablation should be evacuated using high volume suction.
- Equipment should be checked regularly.
- The operator should take adequate precautions to prevent injury/damage to adjacent soft and hard tissue. (Beam/optical components should be used at a reduced power.)
- Provision of interlock to terminate/minimize beam to maximal permissible level. (Interlock is connected to beam shutter, which closes as and when permissible limit exceeds.)
- Provision of warning system (audible tone or flash light), which gets activated as the floor of laser exceeds maximal permissible limit.

Safety measures: Points to remember

- Install laser warning signal.
- Masks, gowns, glove, etc. should be worn during laser procedures
- Wear protective glasses: Dark green for argon and YAG lasers; and clear for CO_2 laser.
- Use wet gauze pads around the target areas (surrounding the treated area).
- Mirrors and other reflecting instruments should be covered; may help stray beams.
- Anesthetic gases should not be placed in the operatory; could be ignited by a laser beam.
- The vaporized soft tissues must be suctioned off; laser plume emitted during laser procedure can carry viruses. Allergy to laser plume has also been reported.

The American Society for Laser Medicine and Surgery have suggested guidelines concerning hazards of laser plume:

- All laser personnel should consider the laser plume potentially hazardous both in terms of the infectivity and allergenicity.
- Evacuator suction system with high-flow volume should be used to collect the plume.
- Eye protection, masks, gloves, and gowns should be worn during laser use; the eyewear protects from splatter, the mask has effective filtration, and the gloves should protect from microbes.

Wearing a surgical mask is essential for infection control; however, the standard surgical mask does not filter out particles <0.5 µm.

A laser surgical mask is now available that will filter to 0.1 µm particles. There are also evacuation systems with filtration for submicron particles that will increase the safety of laser use for biohazardous waste.

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